Sheng-Kai (Kyle) Wen

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WORK EXPERIENCE

Founder & AI Engineer – PuckAI – Hockey Intelligence Platform *May 2025 - Present*
Amherst, MA

- Built and launched PuckAI as a live full-stack hockey analytics product (FastAPI, Next.js, Supabase), giving scouts and fans one place to ask questions about 6,184 NHL skaters and 54,933 scored prospects.
- Shipped a production Retrieval-Augmented Generation (RAG) chatbot over 7,643 scouting reports, combining vector and keyword search (pgvector HNSW, Postgres full-text) with Voyage AI reranking.
- Created a 78-query retrieval benchmark to drive every tuning decision, improving ranking 42% (NDCG@10 0.482 to 0.686); the biggest gain came from routing team and league questions through SQL filters after finding that embeddings could not capture current team membership.
- Moved numeric generation into the data pipeline and added a deterministic validator after finding that prompt-only grounding let the LLM invent or reuse outdated statistics; all 7,643 reports passed, 0 quarantined.
- Traced answers about the wrong player to five separate name-matching leaks and closed them, cutting those answers from 9 to 0 and raising the clean-answer rate from 67% to 80% on a 132-question benchmark.
- Extended the chatbot into a LangGraph agent with web search and highlight videos that chose the right tools on all 26 test questions, fact-checked every web-informed answer, and rejected all 14 wrong-player videos.

AI Researcher (Graduate Extern) – Honda Research Institute (HRI) & 99P Labs *Feb 2026 - May 2026*
Amherst, MA

- Developed a code localization system on DeepCodeBench that ranked the correct file first for 53.5% of queries with Qwen3-32B, ahead of BM25 and an adapted LocAgent baseline (Acc@1 0.535 vs 0.454 and 0.423).
- Built a knowledge-graph retrieval system combining BM25 with intent-based graph traversal, using an AST-based pipeline to index functions, classes, and call/import relationships, plus a test-linkage layer that traced failing tests to their source.
- Designed the evaluation (Fact Recall, Acc@K, LLM-as-judge) and ran ablations across Llama-3.1-8B, Qwen3-8B, and Qwen3-32B on DeepCodeBench (1,144 QA pairs) and LocBench (560 tasks); co-authored the 14-page research report.

Co-Founder & Full-Stack Engineer – Fan2see *Apr 2021 - Apr 2022*
Taipei, Taiwan

- Launched a full-stack PERN product (React, Node.js, Express, PostgreSQL) end-to-end, reaching hundreds of active users with 30% month-over-month growth.
- Cut media-processing latency 25% by building an asynchronous AWS Lambda/S3 pipeline that moved high-volume media transforms off the request path.

APPLIED AI PROJECTS

MAIN-RAG: Multi-Agent Filtering for RAG (UMass Information Retrieval) *Sep 2025 - Dec 2025*

- Reimplemented a multi-stage RAG filtering pipeline (predictor, judge, generator) from scratch with a judge scoring each (query, document, answer) tuple by logit margin, then replaced the LLM judge with rule-based adaptive thresholds (Otsu's method, quantile, mean/std) that matched filtering quality across four datasets.
- Raised how often the kept documents contained the answer from 7.1% to 24.3% (GiP, HotpotQA with Mistral-7B) while cutting retained context from 20 to $\sim$11 documents.

EDUCATION

University of Massachusetts Amherst, M.S. in Computer Science, GPA 3.83/4.00 *Sep 2024 - May 2026*
Coursework: Machine Learning, Natural Language Processing (NLP), Information Retrieval, Artificial Intelligence
National Tsing Hua University, M.S. in Physics *Sep 2020 - Jun 2023*
National Dong Hwa University, B.S. in Electrical Engineering, Minor in Computer Science *Sep 2016 - Jun 2020*

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL, Java, C/C++
- **AI/ML:** RAG, AI agents (LangGraph, Model Context Protocol), LLMs (Claude, GPT-5 mini, Gemini, Qwen3, Llama-3.1), prompt engineering, grounded generation & output validation, LLM-as-judge, fine-tuning (Hugging Face Transformers)
- **Retrieval/Search:** hybrid dense + lexical retrieval, vector search (pgvector, HNSW, FAISS), embeddings & reranking (Voyage AI, Sentence-BERT), BM25, Reciprocal Rank Fusion, Postgres full-text search, knowledge graphs, NDCG / MRR / Recall@K
- **ML/Data:** XGBoost, PyTorch, Pandas, NumPy, feature engineering, model evaluation, ETL pipelines
- **Backend/Frontend/Cloud:** FastAPI, REST APIs, PostgreSQL, Supabase, Next.js, React, Node.js, Docker, AWS (Lambda, S3, EC2), GitHub Actions, Git, Linux, pytest, OpenTelemetry